

SAMARTHA H M

Koppa, Chikkamagaluru, Karnataka

+91 8073339032 | samarth.h.m.gadikal@gmail.com | linkedin.com/in/samarthahm | github.com/samartha-hm

Summary

ECE graduate skilled in PCB design, embedded firmware, and UART/I2C protocols. Built and shipped complete IoT products from R&D through prototyping to market during a startup internship. Comfortable with hardware debugging, sensor integration, and microcontroller programming. Looking for embedded systems, robotics, or IoT roles.

Experience

Experimind Labs Pvt. Ltd.

Feb 2026 – Present

Core Team Member (Promoted from Electronics Intern within 2 months)

- Work across R&D, execution, and intern mentoring - spanning electronics, product development, and cross-functional delivery on multiple company products.
- Engineered Kit 1 (14-component sensor board: 11 input sensors + 3 actuators) and Kit 2 (multi-experiment robotics kit with IR/LDR sensing and motor driver circuitry supporting 9 hands-on activities including line following, light following, and obstacle avoidance) entirely from scratch.
- Developed embedded firmware for Raspberry Pi Pico W autonomous land survey robot — UART communication, GPS/magnetometer/gyroscope sensor fusion, dual-mode navigation — prototype built, debugged, and demonstrated.

Water Quality Monitor | *Product — R&D & System Architecture* | [GitHub](#)

July 2026

- Engineered 4G/LTE cellular connectivity (SIM7670C modem, UART AT commands) and complete AWS cloud pipeline for an ESP32-based multi-parameter water quality system (pH, TDS, turbidity, temperature, dissolved oxygen).
- Built MQTT telemetry over cellular, FastAPI + WebSocket backend, and live dashboard on AWS EC2; solved pH/TDS probe crosstalk using GPIO-controlled sensor phase multiplexing for clean ADC readings.

ISPARK Learning Solutions

Feb – May 2026

Embedded Systems & IoT Intern (Remote)

- Built AI-powered smart waste segregation prototype in 5 days: ESP32 (GPIO sensor interfacing + PWM servo actuation) + ESP32-CAM + Python MobileNetV2 (79% accuracy, 600 images) over custom UART protocol — sub-second DRY/WET/SKIP sorting with real-time telemetry dashboard.

Projects

Autonomous Arecanut Drying Robot | *Capstone Project — Team Lead* | [GitHub](#)

- Fabricated 5-motor 4WD platform with dual-controller architecture — Raspberry Pi 5 (orchestration, Wi-Fi AP, web server) + Arduino Mega (real-time PWM motor control) — over custom 6-byte binary UART protocol achieving <50ms latency and precise positioning accuracy.
- Designed and validated BTS7960 H-bridge drive circuitry (40A peak, thermal stability); debugged hardware-firmware integration using multimeter and oscilloscope; implemented serpentine traversal algorithm for complete field coverage.
- Built complete system spanning Embedded C firmware (I2C/UART/PWM), ML inference pipeline, and React web dashboard with dark mode and hold-to-move safety — field-tested on agricultural terrain. Led 4-member team from design through demonstration.

Personal Projects & Automation Tools | *Independent Builds — [github.com/simply1git](#)*

- Shipped: SyncPadIO (real-time text sharing, deployed), WebsiteON (serverless monitor + Telegram alerts), SonicSync (WebRTC audio), YotuDrive (Reed-Solomon encoding), Mobtola Pro (ADB backup) — Python, React, Node.js, TypeScript.

Technical Skills

Microcontrollers & Platforms: Arduino (UNO/Mega), Raspberry Pi (5/Pico W), ESP32/ESP32-CAM

Protocols & Interfaces: UART, I2C, PWM, GPIO, Wi-Fi, MQTT, 4G/LTE Cellular, Custom Binary Protocols

Cloud & Backend: AWS EC2, FastAPI, WebSockets, REST APIs, SQLite

Firmware & Programming: Embedded C, C/C++, Python, JavaScript/TypeScript, Arduino IDE

Circuit Design & PCB: KiCad, EasyEDA, LTspice, BTS7960 Drivers, SMD/Through-Hole Soldering, PCB Layout & Fabrication

Debugging & Testing: Multimeter, Oscilloscope, Hardware-Firmware Debugging, Sensor Calibration

AI & IoT: MobileNetV2, OpenCV, Edge AI, IoT System Integration, Real-Time Data Acquisition

Tools: Git, VS Code, Linux, Adobe Premiere Pro, Canva, Microsoft Office

Education

Srinivas Institute of Technology, Valachil, Mangalore

Jun 2026

B.E. Electronics & Communication Engineering – CGPA: 7.86/10 (VTU)

Leadership & Extracurriculars

Event Organizer | *ENVISION 2k25 National Tech Fest*

- Organized Line Follower robotics competition for 100+ participants; led promotional design team.

Editor & Designer | *STEADY Department Club (2024 & 2025)*

- Designed 50-page INCHARA magazine distributed to 500+ students and faculty; organized 50+ technical and non-technical events.

Junior Mentor | *TECHBOTS Robotics Club*

- Mentored juniors in embedded programming and PCB design; contributed to robotics systems development.

Content Creator | *DreamBeatz YouTube — 3K+ Subscribers*

- Music and lyrical edit content demonstrating video editing, graphic design, and audience-building ability.